AI651 Deep Learning for Space, Time and Graphs
Assignment 2

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GitHub repository: https://github.com/ahmedjawedaj/dl4stg-pa2

Question 1: River forecasting proposal. Question 2: Motion or memory? (ConvLSTM on Moving MNIST).

Question 1: River Forecasting Proposal

1. Architecture diagram

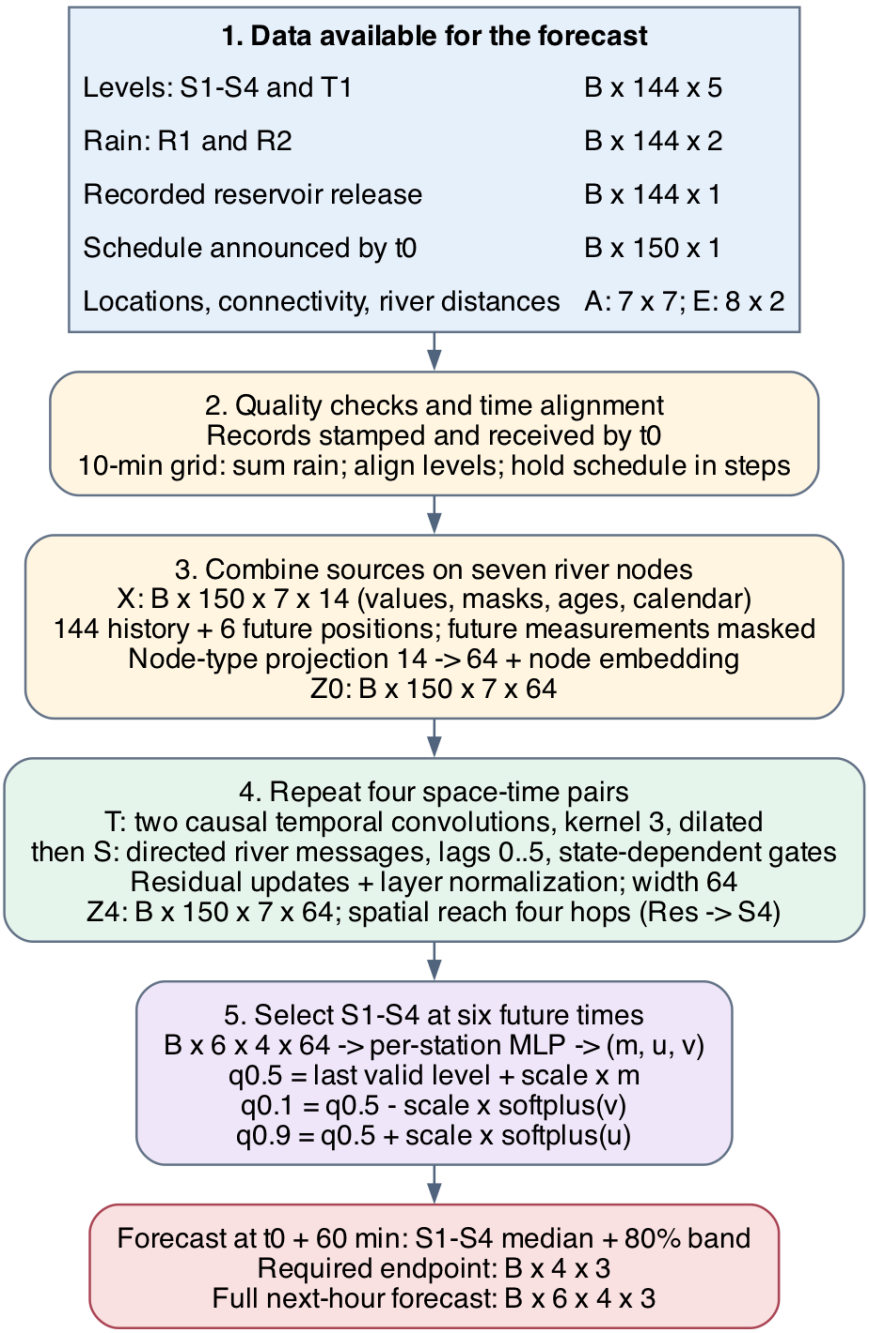


Figure Q1.1: Processing pipeline with main tensor dimensions. B is the batch of issuance times (B = 1 in operation), 150 is the time axis (144 history steps plus 6 known-future steps), 7 is the number of graph nodes and F = 14 is the number of input channels per node and time.

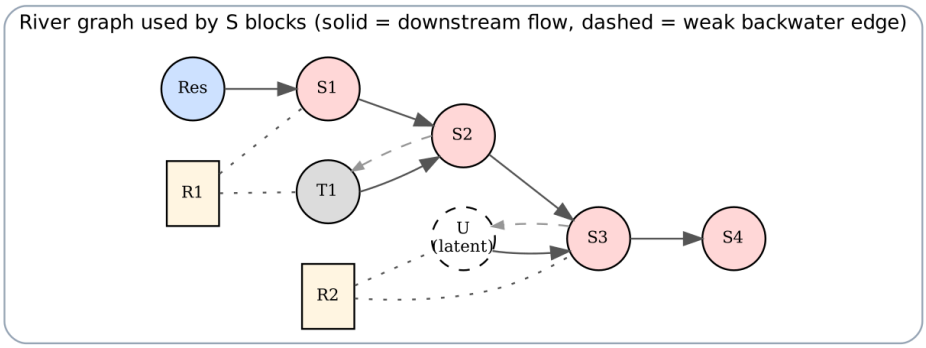


Figure Q1.2: River graph used by the S blocks. Solid arrows follow downstream flow. Dashed arrows are the two weak reverse (backwater) edges at the confluences. U is a latent node for the unmonitored tributary: it has no measurements and receives rainfall only. Dotted links show the gauges' main catchment association. Both gauge series are given to every node and the model learns how much each node uses them.

Proposed system. A river-graph space-time network. Every ten minutes it reads everything known at issuance time t_0 (24 h of levels at S1-S4 and T1, rainfall at R1 and R2, recorded releases, and the release schedule announced up to t_0), places these on a small directed graph that follows the river, and predicts six future levels at S1-S4 ($t_0 + 10, \ldots, t_0 + 60$ min), each with a median and a nominal 80% band. The last horizon is the required one-hour-ahead forecast. Temporal blocks (T) learn how each location responds over time. Spatial blocks (S) pass information downstream along the river, one hop per block. Four T-S pairs give the reach needed for the reservoir signal to arrive at S4 and for travel times of several hours to fit inside the model's view.

Combining sources. All sources are resampled onto one 10-minute grid (rainfall summed over each interval, levels taken at the interval end, schedules held as step functions). Each source becomes channels of the node it describes: levels on S1-S4 and T1, releases and schedule on the reservoir node Res, and rainfall on every node. Channels that do not apply to a node are zero with a zero mask. A projection per node type ($W_{\text{type}}: \mathbb{R}^{14} \to \mathbb{R}^{64}$) maps these heterogeneous inputs to a common width, so a level sensor, the reservoir and the latent tributary node can share the same T and S blocks afterwards. The time axis is extended by six future positions ($t_0 + 10$ to $t_0 + 60$ min). There, only the announced release schedule is filled in. Every measured channel is masked, so future rainfall and future levels never enter the model.

2. Assumptions

Supplied facts (not assumptions). Historical levels at S1-S4 and T1 recorded every 10 minutes. Rainfall at R1 and R2 with timestamps and locations. Recorded releases and announced schedules with announcement and effective times. Station locations, river connectivity and distances. The monitored tributary joins between S1 and

S2 and the unmonitored one between S2 and S3. Precise travel times are not supplied and can change with conditions. Future rainfall and levels are unavailable at issuance.

ID	Assumption	Why reasonable	Effect on design
A1	All timestamps can be converted to one clock (UTC). A level stamp refers to the end of its 10-minute interval.	Telemetry systems normally log in UTC or a fixed local offset.	One shared grid. Rainfall is summed over $(t-10, t]$ so it never includes later rain.
A2	At least 3 years of 10-minute history exist and include several high-flow events.	Typical for an authority that runs a reservoir and flood stations.	Enough data to learn travel times and rainfall response. With less, I would pretrain on synthetic events from a simple routing model (Section 6).
A3	Each record has an arrival time, or the typical reporting delay is known. Most data arrive within 20 minutes, with occasional outages of hours.	Common for radio or cellular telemetry.	Training samples are built "as of" t_0 and delays are simulated. Age channels tell the model how stale each value is.
A4	Level sensors are accurate to a few cm but have spikes, flatlines and gaps. Rain gauges under-catch in strong wind.	Usual pressure and radar level sensors and tipping-bucket gauges.	A rule-based QC step turns bad values into missing values. The model is trained with missing inputs.
A5	Water travels mainly downstream. Backwater (a downstream level raising an upstream one) matters only near confluences.	Downstream dominance follows the supplied flow directions. Restricting backwater to confluences is a modelling assumption to check against observed events.	Graph edges point downstream. A weak reverse edge is kept only across the two confluences.
A6	Travel time from the reservoir to S4 is at most about 8 h, and rainfall reaches the river within about 12 h.	These are provisional routing bounds, not distances or slopes inferred from the schematic. Check them using release-wave timing and rainfall-level lags in the historical record.	Start with a 24-hour history window (144 steps). Extend the window and temporal reach if historical responses exceed these bounds.
A7	Travel time often falls as flow rises, within the historical regime.	A provisional routing simplification to check from observed lag-stage relationships, not a universal law.	State-dependent lag gates avoid enforcing a fixed delay or a strictly monotonic response.
A8	Releases change in steps, are announced before they take effect, and the recorded release is close to the schedule except during operational changes or emergencies.	How release schedules are usually published.	The schedule is a step-function channel. The past difference between actual and scheduled release is an input (Section 5).
A9	Rain at R2 is the best available proxy for the unmonitored tributary catchment, and R1 for the S1 and T1 area.	Gauge positions in the handout figure.	The latent node U gets rainfall but is never trained against a measured value. Both gauges still reach every node.
A10	Channel geometry, station datums and operating rules stay roughly constant over the training period.	Major changes are rare and logged.	One model for all seasons. Retraining after logged changes or detected drift.
A11	Each station's level is in its own local datum.	Usual practice.	Levels are normalised per station and the model predicts the change from the last valid level.
A12	Each rain reading is an accumulation since its preceding timestamp. Within an interval spanning grid boundaries, intensity is treated as uniform.	The handout does not define rainfall units or accumulation intervals. This is a provisional interpretation to verify from gauge metadata.	Allocate interval totals to 10-minute bins by overlap duration, preserving total rain. If readings are rates, integrate them instead.

3. Forecasting task and data specification

- **Forecast origin and information cutoff** t_0: the nominal 10-minute issuance mark (hh:00, hh:10, …). Freeze the input snapshot at t_0: measurements must be both timestamped and received by t_0, and release plans must be announced and received by then. Computation follows immediately, with publication targeted within three minutes and the as-of time retained. Later arrivals cannot enter this original forecast.
- **History window**: 144 steps, $t_0 - 1430$ min to t_0 (24 h).
- **Horizon**: the next hour in six 10-minute steps. The $t_0 + 60$ endpoint satisfies the handout, while all six steps support the tutorial's next-hour display.
- **Outputs**: for S1, S2, S3 and S4, the median and the 10% and 90% quantiles at each horizon. The 10%-90% interval is a nominal 80% band whose coverage must be validated.

Tensor	Dimensions	Axis meanings	Timestamps / coverage
Levels L (+ mask, + age)	B x 144 x 5	issuance sample, history step, station (S1, S2, S3, S4, T1)	$t_0 - 1430 \dots t_0$ min, value at step end
Rainfall P (+ mask)	B x 144 x 2	sample, history step, gauge (R1, R2)	accumulation over $(t-10, t]$
Recorded release Q^{act} (+ mask)	B x 144 x 1	sample, history step, reservoir	$t_0 - 1430 \dots t_0$
Announced schedule Q^{sch} (+ announcement age)	B x 150 x 1	sample, history + future step, reservoir	$t_0 - 1430 \dots t_0 + 60$. Only announcements received by t_0, with past positions reconstructed as of their own time
Calendar	B x 150 x 2	sample, step, (sin, cos) of time of day	$t_0 - 1430 \dots t_0 + 60$ (always known)
Graph A, edge features E	7 x 7, 8 edges x 2 (6 downstream + 2 weak reverse at the confluences)	node, node / edge, (distance along river, direction)	static
Assembled input X	B x 150 x 7 x 14	sample, step, node (Res, S1, T1, S2, U, S3, S4), channel	history plus the six future steps (measured channels masked there)
Required endpoint $\widehat{Y}$	B x 4 x 3	sample, station (S1-S4), quantile (0.1, 0.5, 0.9)	$t_0 + 60$ min
Full forecast $\widehat{Y}^{\text{all}}$, training target Y	B x 6 x 4 x 3, B x 6 x 4	sample, horizon, station, (quantile)	$t_0 + 10 \dots t_0 + 60$ min

The 14 channels per node and step are: level value, level mask, level age, R1, R2, R1 mask, R2 mask, recorded release, release mask, scheduled release, schedule age, future flag, and time of day (sin, cos).

Alignment. *Across time:* every source is placed on the same 10-minute grid ending at t_0 (A1). Rain totals with irregular stamps are allocated to bins under A12, using only records already received. A value that is missing or has not arrived is set to the last valid value with mask 0. Level age is recorded explicitly in minutes. For rainfall and actual release, the sequence of masks identifies gaps within the window, without an additional age channel. If there is no previous valid value, use zero with mask 0. *Across locations:* stations are not interpolated onto a regular spatial grid. Each location is a graph node and distance enters as an edge feature, so the uneven spacing is kept. *Available at issuance:* everything in the history columns that arrived by t_0, the full release

schedule as announced up to t_0, the calendar and the static graph. Not available: rainfall and levels after t_0, schedule changes announced after t_0, and the actual releases after t_0.

4. Model computations and design justification

Notation. $\tau \in \{t_0 - 143\Delta, \ldots, t_0 + 6\Delta\}$ indexes the 150 steps (Δ = 10 min). $i \in V$ indexes nodes. $x_i(\tau) \in \mathbb{R}^{14}$ is the input. $z_i^{(\ell)}(\tau) \in \mathbb{R}^{64}$ is the representation after block ℓ. $\mathcal{N}(i)$ contains incoming graph neighbours, including the two reverse edges in Figure Q1.2. r_{ji} labels an edge as downstream or reverse, and e_{ji} contains scaled river distance and direction. Inputs outside the finite time window are zero-padded. Block parameters are learned separately for each of the four pairs.

Input projection. $z_i^{(0)}(\tau) = W_{\text{type}(i)}\, x_i(\tau) + b_{\text{type}(i)} + \epsilon_i$, where ϵ_i is a learned node embedding.

T block. Each pair applies two residual causal convolutions independently at each node. With $v_i^{(0)} = z_i$ and $a = 1, 2$,

$$v_i^{(a)}(\tau) = v_i^{(a-1)}(\tau) + GELU\left(LN\left(b^{(a)} + \sum_{k=0}^{2} W_k^{(a)} v_i^{(a-1)}(\tau - k d_a)\right)\right), \qquad u_i = v_i^{(2)}.$$

Each $W_k^{(a)}$ is 64×64. LN normalises the 64 channels at one node and time. Dilations are (1, 2), (4, 8), (16, 32), (64, 1) across the four pairs, giving $1 + 2(128) = 257$ temporal positions. No input after τ is used.

S block. A state-dependent gate controls each incoming edge and lag:

$$g_{ji\delta}(\tau) = \text{sigmoid}\left(a_{\delta, r_{ji}}^{\top}[u_i(\tau), u_j(\tau - \delta), e_{ji}] + b_{\delta, r_{ji}}\right),$$

$$m_i(\tau) = \sum_{j \in \mathcal{N}(i)} \beta_{r_{ji}} \sum_{\delta=0}^{5} g_{ji\delta}(\tau) U_{\delta, r_{ji}} u_j(\tau - \delta),$$

$$z_i^{+}(\tau) = u_i(\tau) + GELU\Big(LN\big(W_s u_i(\tau) + m_i(\tau) + b_s\big)\Big).$$

$U_{\delta, r}$ and W_s are 64×64. Downstream and reverse relations use separate weights. Set $\beta_{down} = 1$ and $\beta_{reverse} = 0.1$ as a design prior for A5, not a measured physical ratio or a bound on learned message magnitude. The six lag taps cover 0-50 minutes per S block. Their gates depend on the current destination state and lagged source state, allowing different delays to be emphasised under different conditions. These are local sigmoid gates, not global attention.

Reach. Each S block moves information one hop. The longest path the forecast needs is Res to S1 to S2 to S3 to S4, four hops, so four S blocks are the minimum and are used. Temporal reach is the T-block reach plus up to 4×5 lag steps from the S blocks, so any input in the window at a node upstream of a station (or at the station itself) can affect that station's forecast. Information does not travel upstream, except across the two weak confluence edges, consistent with the modelling assumption A5.

Forecast mapping. For station $s \in$ {S1, …, S4} and horizon $k \in \{1, \ldots, 6\}$, with $\ell_s(t_0)$ the last valid level and σ_s the station's training standard deviation of 1-hour changes:

$$[m, u, v]^{\top} = W_{s,2}\, GELU\Big(W_{s,1} z_s^{(4)}(t_0 + k\Delta) + b_{s,1}\Big) + b_{s,2}, \qquad \widehat{q}_{0.5} = \ell_s(t_0) + \sigma_s m,$$

$$\widehat{q}_{0.9} = \widehat{q}_{0.5} + \sigma_s\, softplus(u), \qquad \widehat{q}_{0.1} = \widehat{q}_{0.5} - \sigma_s\, softplus(v).$$

$W_{s,1}$ is 32×64 and $W_{s,2}$ is 3×32, shared across horizons within station s. A zero median increment gives persistence, and the softplus offsets keep the quantiles ordered. All normalisation statistics are fitted on the training period only.

Training objective. Pinball loss over valid targets, with the one-hour horizon weighted most:

$$\mathcal{L} = \frac{1}{|\Omega|} \sum_{(b,k,s) \in \Omega} w_k \sum_{q \in \{0.1, 0.5, 0.9\}} \rho_q(y_{bks} - \widehat{q}_{q,bks}), \qquad \rho_q(r) = \max(qr, (q-1)r),$$

with $w_6 = 1$, $w_{1..5} = 0.3$ and Ω the samples, horizons and stations whose target is a valid measurement. Training samples are every 10-minute issuance time in the training years, built as of t_0 (A3). During training, random sensor gaps (10 min to 6 h, any node) and simulated reporting delays are applied so the model learns to forecast with incomplete data. Adam, early stopping on a later validation period, and a final test on the most

recent period. Data are split by time, never at random, with flood-event coverage checked in each period rather than assumed. If extremes are absent from the test period, extreme-event accuracy cannot be established.

How each source reaches each forecast.

- **S1**: its own history, the reservoir node one hop upstream (recorded releases and the announced schedule for the next hour), and rainfall (mostly R1).
- **S2**: S1 and T1 one hop upstream (main river plus monitored tributary), the reservoir two hops upstream, and rainfall.
- **S3**: S2 one hop upstream, the latent node U (driven by R2 rainfall), and everything upstream of S2 within three hops.
- **S4**: S3 one hop upstream, then everything above it, reaching the reservoir in four hops.

Every recorded release in the 24-hour window can reach S4's forecast, because four hops cover the path and the window is longer than the assumed travel time (A6). This does not mean a release made at t_0 changes S4 within the hour. Physically it only reaches stations whose travel time from the reservoir is under one hour, and the model has to learn which stations those are.

Physical assumptions versus learned mappings. Physical structure is built in: the graph follows the river and its direction (A5), the T blocks are causal, distance is an edge feature, the forecast is a change from the last observed level, and no future measurement can enter. Learned from data: travel times and how they change with flow (A7), attenuation of waves, rainfall-runoff response, the unmonitored tributary's contribution, sensor biases and the uncertainty width. Mass balance is not enforced, because levels (not discharges) are measured and rating curves are not supplied.

Why this design. A temporal-only model cannot use upstream signals that have not yet reached a station, and a spatial-only model cannot represent travel delays. Alternating T and S blocks gives both, with a reach that is easy to check against the river. Attention would also connect distant stations, but with only seven nodes and a known physical topology the graph already provides every path needed, at lower cost, with a structure reviewers can audit. A purely physical hydraulic model would need cross-sections, roughness and the unmonitored inflow, none of which are supplied. Quantile outputs give the committee an honest band instead of a single number.

5. Operation and exceptional conditions

(i) Missing observations. QC rules (physical range, rate-of-change limit, flatline longer than 1 h) convert bad values to missing. A missing value is replaced by the last valid one with mask 0 and its age. Because training randomly hides inputs, the model learns to lean on neighbours (for example, S3 from S2 and U when S3 is down) and can learn wider bands when inputs are stale. Widening is not guaranteed by the head: check empirical coverage under outages and recalibrate or flag undercoverage. If a station's own level has been missing for over 6 h, its forecast is still issued (its last valid level is old) but flagged as "degraded", and the band comes from the mask-conditioned quantile head. If all inputs for a forecast are missing, the service issues no forecast for that station rather than a fabricated one.

(ii) Delayed observations. Each forecast is built from a data snapshot taken at the cut-off. A value that has not arrived is treated as missing for that run (age channel set). When delayed data arrive, any updated forecast for the same target time is labelled with its later as-of time and shorter remaining horizon. It is a revised forecast, not information available to the original t_0 run. The earlier version is kept in the archive, so evaluation always uses the forecasts that were actually issued. Training simulates the observed delay distribution (A3) so the model is used to this situation.

(iii) Announced future releases. The schedule channel holds, for each 10-minute step from $t_0 - 1430$ to $t_0 + 60$ min, the release that was announced to be in force at that step, using only announcements available by the historical step for past positions, or by t_0 for future positions. When several announcements cover the same step, the latest eligible announcement wins. A plan announced after t_0 is excluded from this run. The schedule-age channel tells the model how recently it was confirmed. Because the six future positions carry the schedule,

a release change starting at $t_0 + 20$ min can affect S1's forecast if the model has learned that releases reach S1 within the remaining 40 min. Downstream stations are affected only through their upstream path, as in reality.

(iv) Scheduled versus actual releases. The past uses recorded actual releases (Q^{act}). The past schedule as it was known at each time is a separate channel, so the model sees how far actual releases have drifted from schedules and learns how much to trust the next hour's schedule. Operationally, if the latest recorded release differs from the schedule by more than a set threshold (for example 10%), the system re-runs immediately and issues the forecast under the current actual release held constant for the next hour, alongside the schedule-based forecast, and flags the discrepancy to operators. An unannounced release cannot be detected from its release record before that record arrives. Re-running on receipt gives a typical delay of up to about 20 minutes under A3, plus processing, rather than a guaranteed response within one 10-minute cycle.

Feasibility of running every ten minutes. The model has roughly 0.3 M parameters. One forecast applies the T and S blocks at 150 x 7 positions, about 2.5×10^5 multiply-adds per position, so roughly 3×10^8 in total. Even a conservative sustained rate of 10^8 multiply-adds per second gives about three seconds of arithmetic, leaving substantial room within the 10-minute cycle for software overhead and data handling. This is an operation-count estimate, not a measured latency. Data ingestion, QC and assembly are simple table operations. Re-runs for late data or release discrepancies have the same model cost. A small server is a reasonable starting deployment, with latency checked on that server before operation and a persistence forecast published automatically if the model run fails.

6. Uncertainty, limitations, and additional information

Uncertainties and limitations

Issue	Forecast impact	Response, or what remains unresolved
Unmonitored tributary between S2 and S3	Unobserved inflow raises S3 and S4 with no upstream warning. Local rain can cause large S3/S4 errors. Wider bands are desirable but must be checked, not presumed.	Latent node U driven by rainfall (mostly R2) learns this inflow implicitly from what S3 receives beyond the routed S2 signal. Unresolved: the model cannot tell an unseen inflow from a change in S2-to-S3 travel time, and R2 may miss local storms.
Travel times change with conditions	A fixed lag would put peaks too early at low flow and too late at high flow.	Flow-gated lag taps and long temporal reach let the effective lag depend on current stage (A7). Unresolved: floods larger than any in the training record need extrapolation, so their timing is less reliable.
Rare extreme events	The most important forecasts have the least training data. Quantiles may be under-dispersed in the tail.	Event-weighted validation, conformal recalibration of the band per station on recent residuals, optional pretraining on synthetic floods from a simple routing model.
Rainfall representativeness (two point gauges)	Convective storms between gauges are missed. Under-catch biases totals low (A4).	Rain is one input among several. Check band coverage separately during heavy rain and recalibrate if needed. Radar would help (requests).
Sensor drift, datum shifts, channel change after floods	Biased inputs and targets. Learned relationships may become wrong.	QC, drift monitoring of residuals per station, retraining after logged changes (A10).
Release behaviour outside training (new operating rules, emergencies)	Releases unlike any in training can be mis-routed.	Discrepancy re-runs (Section 5 (iv)) and operator flags. Unresolved for completely new release patterns until retrained.
Reporting delays and outages	Stale inputs make forecasts closer to persistence.	Age channels, delay simulation in training, versioned re-issue (Section 5 (ii)).

Additional information requests

Request	How it helps	Usable without it?
Arrival timestamps or delay logs for all telemetry	Realistic delay simulation and an exact as-of reconstruction for training.	Yes, delays are assumed (A3).
Rating curves and surveyed cross-sections at S1-S4 and T1	Convert levels to discharge, add a mass-balance check, estimate wave speeds and travel times.	Yes, the model works on levels.
Historical travel-time studies or release-wave timing	Sets the window and lag ranges, and allows checking the learned lags.	Yes, lags are learned.
A gauge on the unmonitored tributary, even a temporary campaign	Directly constrains node U, the largest uncertainty at S3 and S4.	Yes, with wider bands at S3 and S4.
Rain radar and rainfall forecasts issued before t_0	Better spatial rain and a look-ahead for the next hour. A forecast issued before t_0 is legitimate input.	Yes.
Reservoir level, gate settings and spillway status	Better estimate of the actual release and early warning of unscheduled changes.	Yes.
Station datums, maintenance and recalibration logs	Lets QC separate real changes from instrument shifts.	Yes, but drift detection is slower.

Question 2: Motion or Memory?

All code, checkpoints, logs and GIFs are in the repository linked on page 1. Every number below comes from the scripts named in brackets. Learned-model entries are **mean (seed 0, seed 1)**. Figures and per-video histograms use the seed-0 model unless stated. Errors are given to three significant figures, with scientific notation below 10^{-3}. **Hardware:** Apple M1 Max, 64 GB RAM, macOS-15.6.1-arm64-arm-64bit-Mach-O, PyTorch 2.14.1, device mps.

1. Setup

Data. `torchvision.datasets.MovingMNIST(root="data", split=None, download=True)` gives all 10,000 videos (20 x 1 x 64 x 64, uint8). Videos are split by index: 0-7,999 train, 8,000-8,999 validation, 9,000-9,999 test, and divided by 255 into [0, 1] (`pa2/data.py`). Mirror is `torch.flip(x, dims=(-1,))`, rotation is `torch.rot90(x, k=1, dims=(-2, -1))`, and a group element MR^a applies R^a first and then M. The window function returns the five input frames, the existing target frames (zero-filled beyond frame 20) and a validity mask, and it is the same function used in training and evaluation.

Table 1. Transformation and frame-window checks (`scripts/table1_checks.py`). All rows match.

Check	Expected	Yours
(a) max|M(M(x)) - x|	0	0 (torch.equal=True)
(b) max|R^4(x) - x|	0	0 (torch.equal=True)
(c) max|M R M(x) - R^-1(x)|	0	0 (torch.equal=True)
(d) bright pixel of M(p) (frame, row, column)	(7, 10, 43)	(7, 10, 43)
(e) bright pixel of R(p) (frame, row, column)	(7, 43, 10)	(7, 43, 10)
(f) 1x input / target frames	1-5 / 6-10	1, 2, 3, 4, 5 / 6, 7, 8, 9, 10
(g) 2x input / target frames	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19
(h) 3x input / target frames	1, 4, 7, 10, 13 / 16, 19	1, 4, 7, 10, 13 / 16, 19

Figure 1. Farneback optical flow on the first 50 training videos (950 consecutive frame pairs), keeping pixels brighter than 0.2 in both frames with flow magnitude above 0.5 px (`scripts/fig1_flow.py`).

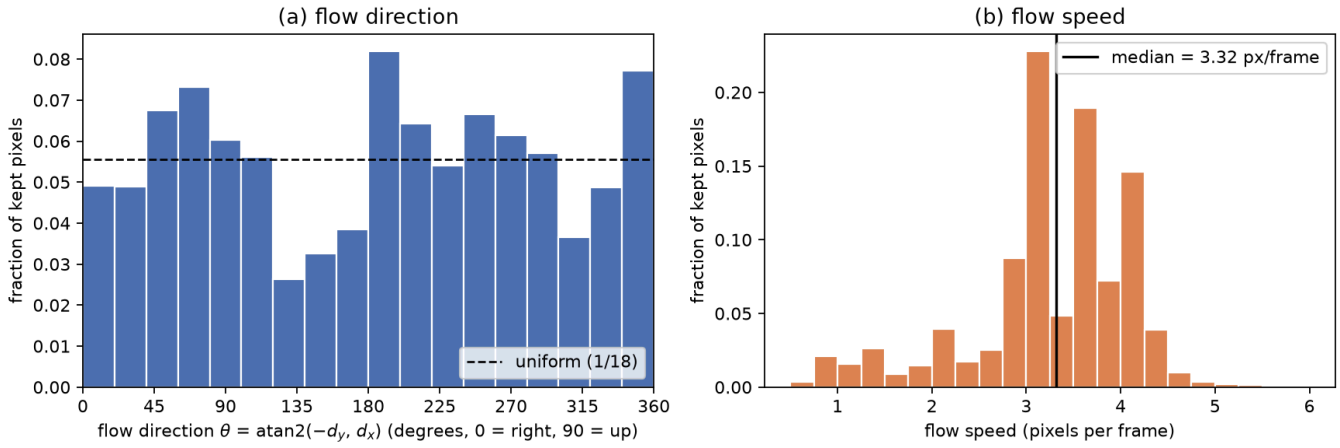


Figure 1. (a) Direction histogram, fraction of kept pixels in 18 bins of 20°, dashed line at 1/18. (b) Speed histogram, fraction of kept pixels, with the median marked. Median speed 3.32 px/frame, 103,050 pixels kept from 950 frame pairs.

Q2.1

> **Direction.** Every 20° bin is populated: the smallest holds 0.0263 of the kept pixels and the largest 0.0820, against 0.0556 for a uniform distribution. The histogram is uneven (about a 3x spread between bins), but no

direction is missing or rare. Mirroring every video would map each direction θ to $180° - \theta$, and a 90° rotation would map θ to $\theta + 90°$. Both only move mass between bins that are all well populated. The L1 distance between the histogram and its mirrored version is 0.264, and 0.400 for its rotated version, the same order as its distance to a uniform histogram (0.220). With only 100 digits in 50 videos, each contributing a few straight segments, this level of unevenness is expected from sampling. So mirrored and rotated **trajectories** should already be familiar to a model trained on this data. Only the mirrored or rotated digit **shapes** are new.

Speed. A typical digit moves 3.32 px per frame (median). 80% of kept pixels lie between 2.14 and 4.12 px/frame and 99% are below 4.67. At 2× a typical digit therefore moves about 6.64 px between the frames the model sees, and at 3× about 9.95 px. Both are far outside the training range: only 0.0340% of kept training pixels move faster than 6 px/frame (the x-axis of Figure 1(b) stops at 6 px).

Overlap bias. Where digits overlap, Farneback sees one merged blob whose shape changes, so flow there is unreliable. I checked how much this matters by splitting the frame pairs with a connected-component test (`n_blobs` in `pa2/data.py`): 60.9% of kept pixels come from pairs where the two digits touch or overlap in at least one frame. Their median speed is 3.30 px/frame against 3.51 for pairs with separate digits, so overlap pulls speeds down only slightly. They also make up 60.4% of the slowest 10% of pixels, about their overall share, so they do not dominate the slow tail. Using only the pairs with separate digits, the direction histogram still covers every bin (0.0257 to 0.0818, L1 distance 0.109 from the full histogram). The bias is too small to change either answer: all directions are familiar, and 2× and 3× are not.

Q2.2

Transformation	Motion statistics	Digit appearance
Mirror M	Unchanged in distribution. Directions map $\theta \to 180° - \theta$, speeds are kept, and bounces stay valid in the symmetric box. Since training directions already cover the circle (Q2.1), mirrored motion is in-distribution, although a finite training set is never exactly symmetric.	Changed: a mirrored 2, 3, 4, 5, 7 or 9 is not a digit shape the model has seen.
Rotation R	Unchanged in distribution, for the same reason ($\theta \to \theta + 90°$, speeds kept).	Changed even more: strokes that are vertical become horizontal, and all ten digit shapes are new.
Speed-up 2×, 3×	Changed: displacements per seen frame double or triple, outside the training range.	Unchanged: every frame is an ordinary training-style frame.

This makes the three tests a controlled experiment. Mirror and rotation hold the motion distribution fixed and change only appearance, so a failure there points to a motion rule tied to how digits look. Speed-up holds appearance fixed and changes only the motion, so a failure there points to unfamiliar motion or to limited reach. These controls suggest hypotheses. The A-to-B and A-to-C-to-D comparisons are needed to distinguish causes.

2. Baseline: ConvLSTM

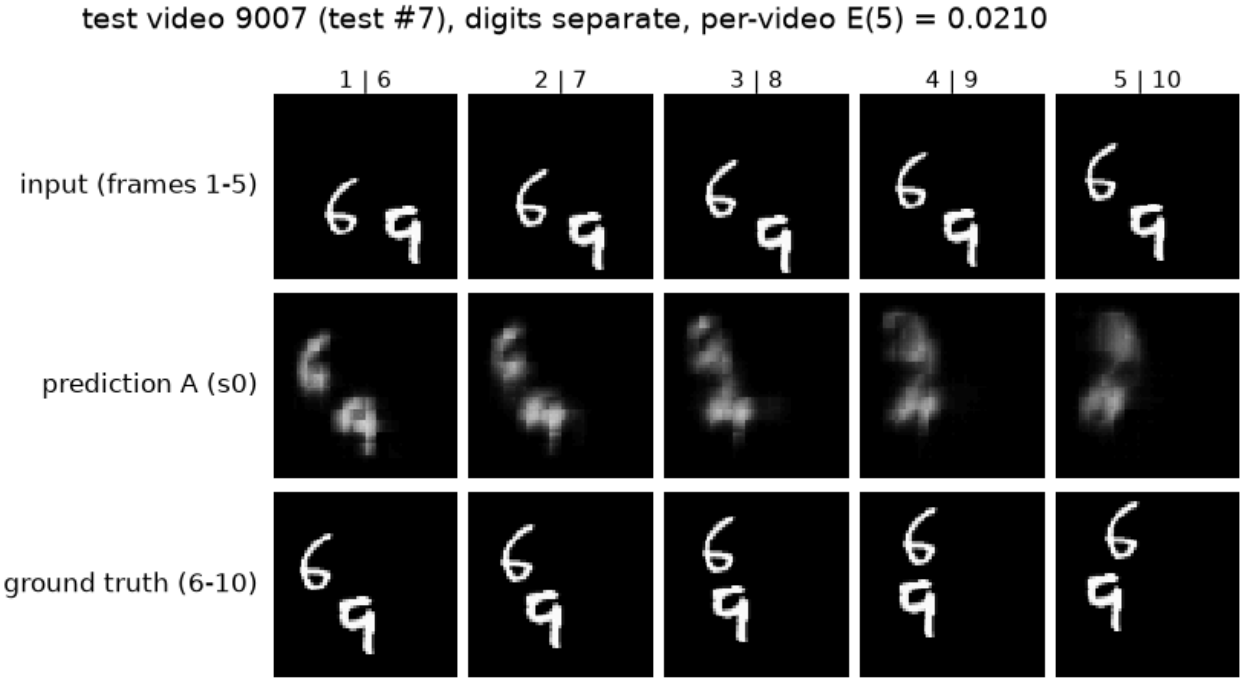
Model A (`pa2/models.py`, config `A`): `pixel_unshuffle(x, 4)` to 16 x 16 x 16, two ConvLSTM layers (64 hidden channels, 3 x 3, stride 1, padding 1, one convolution producing $[i, f, o, g]$, no peepholes, zero initial states), a 1 x 1 convolution 64 to 16, `pixel_shuffle(., 4)` and a sigmoid. PyTorch default initialisation and `bias=True` everywhere. **Trainable parameters: 480,784** (layer 1: $(16 + 64) \cdot 256 \cdot 9 + 256 = 184,576$, layer 2: $128 \cdot 256 \cdot 9 + 256 = 295,168$, output: $64 \cdot 16 + 16 = 1,040$). The forward pass runs 9 steps: steps 1-5 read the input frames, steps 6-9 read the model's own previous prediction (no teacher forcing, gradients flow through the fed-back frames), and the outputs of steps 5-9 form $\widehat{Y}$.

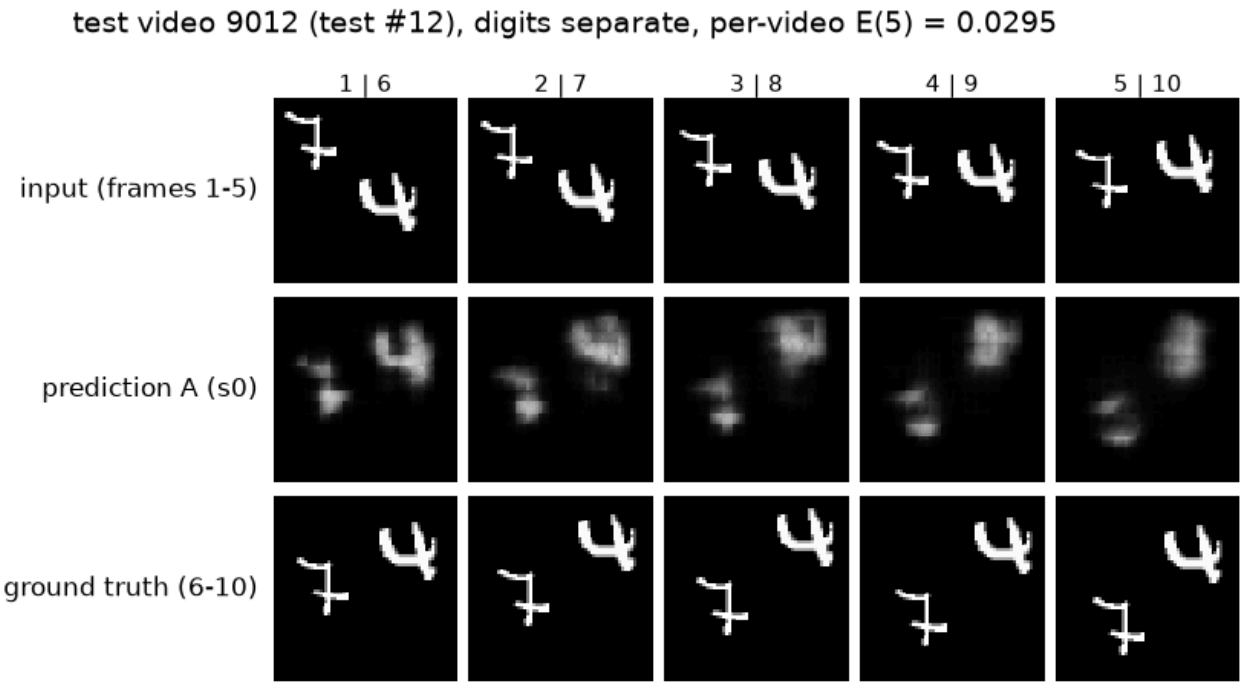
Training (`pa2/train.py`): Adam, learning rate 10^{-3}, no schedule, batch 32, 30 epochs, final-epoch weights kept. Each epoch shuffles the 8,000 training videos and draws one window per video (models A, B: $t \in \{1, ..., 11\}$ at

1×). The loss is the squared error averaged over pixels and valid target frames of each video, then over the videos in the batch. Seeds 0 and 1 control initialisation, shuffling, window and augmentation sampling. Validation (1×, $t = 1$) is logged every epoch for monitoring only.

Table 2. Test errors at 1×, $t = 1$ (`scripts/phase1_baseline.py`).

Model	MSE_1	MSE_2	MSE_3	MSE_4	MSE_5	$E^{(5)}$	Trainable params
A (baseline ConvLSTM)	0.0168 (0.0171, 0.0166)	0.0202 (0.0206, 0.0199)	0.0225 (0.0229, 0.0221)	0.0242 (0.0247, 0.0238)	0.0259 (0.0264, 0.0255)	0.0219 (0.0223, 0.0216)	480,784
Blank	0.0426	0.0426	0.0426	0.0425	0.0425	0.0426	0
Copy-last	0.0492	0.0604	0.0669	0.0708	0.0732	0.0641	0





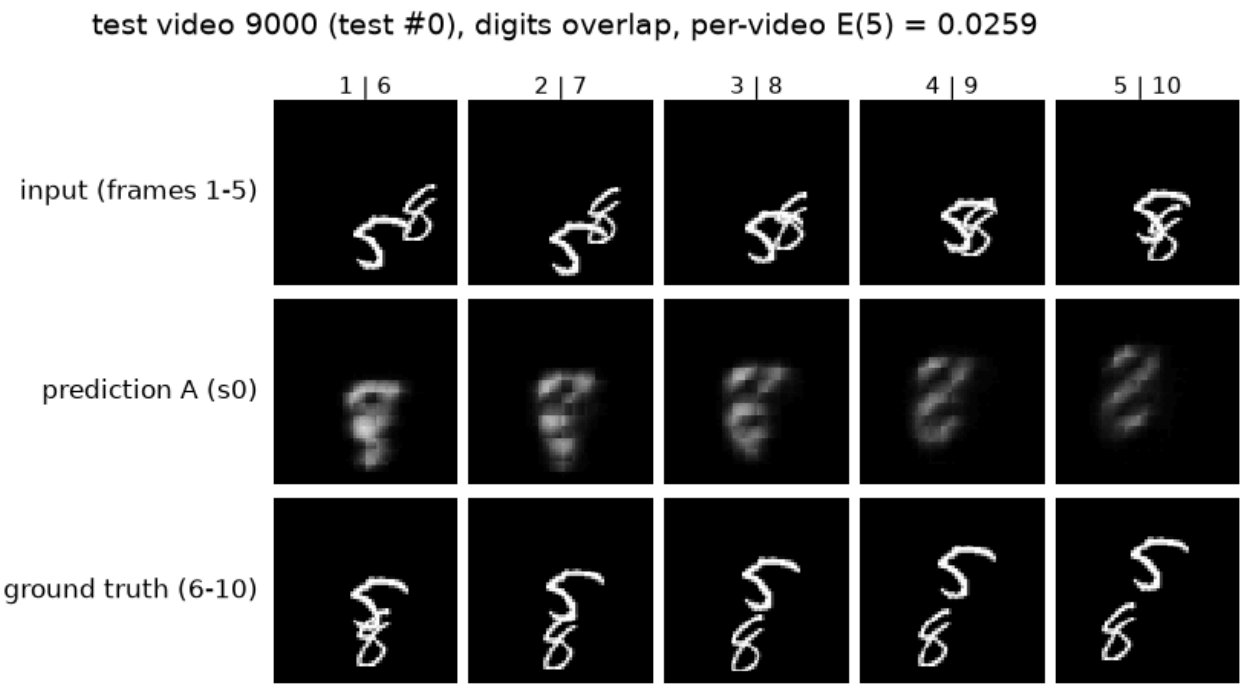


Figure 2. Model A (seed 0) on three test videos at 1×, $t = 1$: input frames 1-5, predictions for frames 6-10, and ground truth frames 6-10. Videos 9007, 9012, 9000 (dataset index). Video 9000 has overlapping digits.

Q2.3

Reach of model A: 8 px per time step.

1. *Patch folding.* `pixel_unshuffle(x, 4)` turns each 4 x 4 pixel patch into one cell of a 16 x 16 grid. Moving one cell means moving 4 original pixels (centre to centre). Folding itself moves nothing (convention).
2. *One 3 x 3 convolution* (stride 1, dilation 1) mixes a cell with its 8 neighbours, so information moves at most **1 cell** per convolution.
3. *Layer 1* at step τ applies one 3 x 3 convolution to $[u_\tau, h^{(1)}_{\tau-1}]$. The gates are pointwise, and the cell update $c_\tau = \sigma(f) \odot c_{\tau-1} + \sigma(i) \odot \tanh(g)$ and $h_\tau = \sigma(o) \odot \tanh(c_\tau)$ are elementwise, so $h^{(1)}_\tau$ can depend on inputs at most 1 cell away.
4. *Layer 2* applies one 3 x 3 convolution to $[h^{(1)}_\tau, h^{(2)}_{\tau-1}]$. Another cell.
5. *Output.* The 1 x 1 convolution, `pixel_shuffle` and sigmoid do not move information.
6. *All paths within one step to $\widehat{x}_{\tau+1}$:* from u_τ: layer-1 conv then layer-2 conv = 2 cells. From $h^{(1)}_{\tau-1}$: the same two convolutions = 2 cells. From $h^{(2)}_{\tau-1}$: only the layer-2 conv = 1 cell. The recurrence does not add more within a step, because each state is used once per step. The longest path is **2 cells = 2 x 4 = 8 original pixels**.

An autograd check (`scripts/reach_check.py`) confirms it: the gradient of one output cell with respect to u_τ and $h^{(1)}_{\tau-1}$ is non-zero only within 2 cells, and with respect to $h^{(2)}_{\tau-1}$ within 1 cell. Over several steps reach adds up (8 px per step), but a digit's new position must connect to its old position, held in the state, within one update. With Figure 1's median of 3.32 px/frame, 1× motion is well inside 8 px, 2× (6.64 px) is near the limit, and 3× (9.95 px) exceeds it.

Q2.4

Table 2 shows mean E(5) = 0.0219 for A (seed 0: 0.0223, seed 1: 0.0216), compared with 0.0426 for Blank and 0.0641 for Copy-last. A reduces error by 48.5% relative to Blank and 65.8% relative to Copy-last. Copy-last becomes increasingly misplaced as the digits move, whereas Blank incurs roughly constant error by predicting the background everywhere. A therefore learns useful predictive structure beyond either reference, although low pixel error alone does not establish a correct motion rule.

A's mean horizon MSE increases from 0.0168 at h = 1 to 0.0259 at h = 5, a 54.0% rise, as errors accumulate through autoregressive feedback. In Figure 2, the separated digits in videos 9007 and 9012 remain approximately localized initially but lose their stroke shapes and become diffuse over the rollout. The 7 and 4 in video 9012 become small blurred patches. In overlapping video 9000, the true digits pass through one another and later separate vertically, while A produces a merged, blurred vertical trace that does not recover two clear digit shapes. These examples qualify the average improvement: A predicts better than the references but is neither sharp nor reliably resolving overlap.

Q2.5 (prediction, written and committed before any test in Section 3)

I predict that model A will not pass exact mirror consistency, and that a 90-degree rotation will produce an even larger inconsistency and transformed prediction error. Figure 1 suggests that motion in all directions is already familiar, but the transformed digit shapes were not part of training, and ordinary convolutions do not enforce either symmetry. Figure 2 already shows blurred, imperfect forecasts at normal speed, so I do not expect correct transformed trajectories merely because the underlying motion rule remains valid.

Using the measured median flow of 3.32 pixels per frame and the 8-pixel one-step reach, I expect 1x motion to be easiest, 2x motion (about 6.64 pixels between observed frames) to degrade, and 3x motion (about 9.95 pixels) to break down most strongly. I expect faster clips to produce diffuse or misplaced digits rather than cleanly extrapolated motion. This is a qualitative prediction, not a sharp speed threshold: maximum structural reach need not be the reach the learned model actually uses, and pixel-flow speed is only an approximate description of a digit's motion.

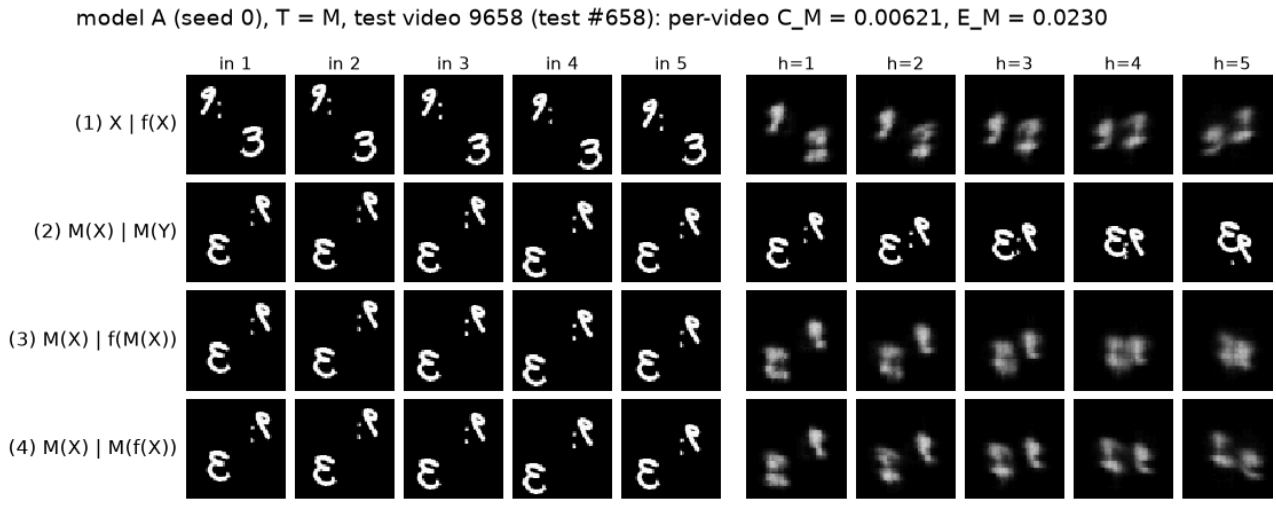
Inserted by the report build directly from the git commit recorded in `logs/phase2_gate.txt` (`answers/Q2_5.md`), committed at 2026-10-19T17:26:31+05:00 (commit 2513a26). The stress tests started at 2026-10-19T17:28:23+0500 (`logs/phase2_gate.txt`).

3. Testing the Baseline

Mirror and rotation

Table 3. Error at 1×, consistency gaps and transformed errors, $t = 1$ (`scripts/phase2_stress.py`).

Model	$E^{(5)}$	C_M	E_M	C_R	E_R
A	0.0219 (0.0223, 0.0216)	0.00178 (0.00172, 0.00183)	0.0226 (0.0229, 0.0222)	0.00280 (0.00271, 0.00290)	0.0236 (0.0238, 0.0234)
Blank	0.0426	0	0.0426	0	0.0426
Copy-last	0.0641	0	0.0641	0	0.0641



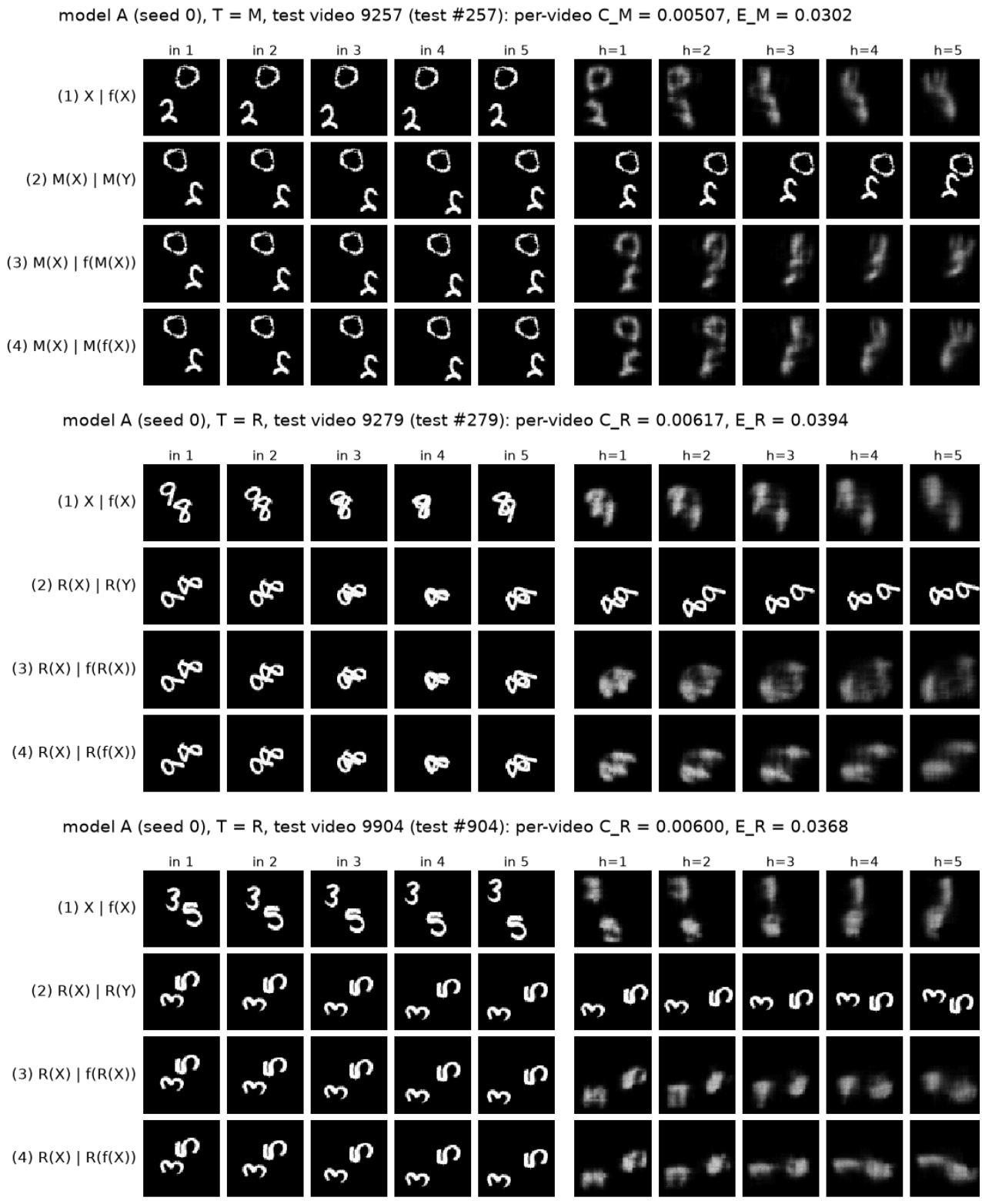


Figure 3. The two test videos with the largest per-video C_M and the two with the largest C_R (model A, seed 0). Rows: (1) X | f(X), (2) T(X) | T(Y), (3) T(X) | f(T(X)), (4) T(X) | T(f(X)). GIFs stepping through the five horizons: `gifs/fig3_A_M_v658.gif`, `gifs/fig3_A_M_v257.gif`, `gifs/fig3_A_R_v279.gif`, `gifs/fig3_A_R_v904.gif`.

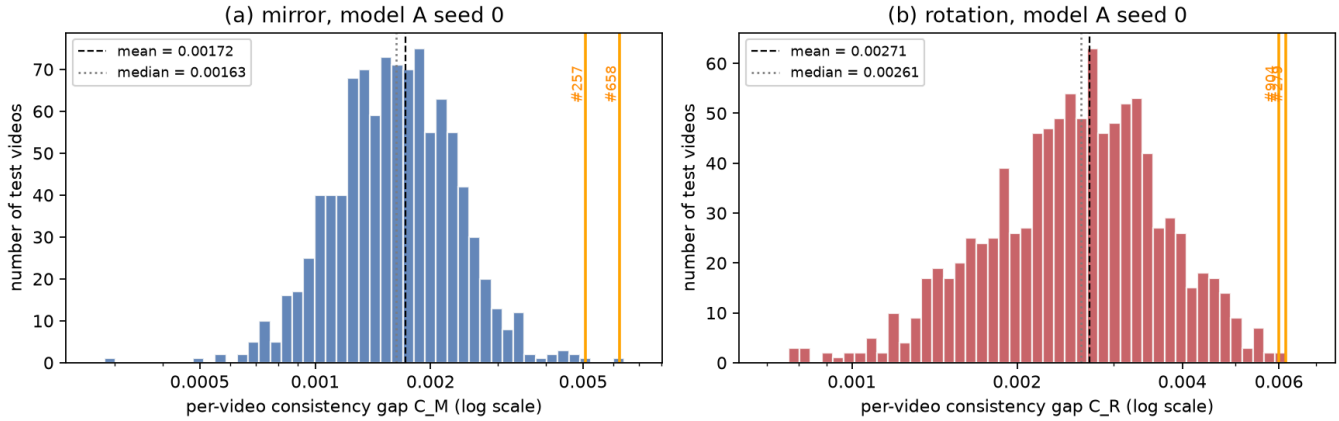


Figure 4. Per-video C_M (a) and C_R (b) of model A (seed 0) over the 1,000 test videos, log scale. Orange lines mark the Figure 3 videos.

Q2.6

> Table 3 gives C_M = C_R = 0 for both Blank and Copy-last, even though their 1x transformed E(5) errors are 0.0426 and 0.0641. Blank commutes with each transformation because transforming an all-zero image changes nothing. Copy-last commutes because selecting the last input frame and repeating it can be done before or after transforming the frames. Thus exact consistency tests whether a predictor respects the transformation, not whether it forecasts motion correctly. E_T must accompany C_T to distinguish an accurate equivariant forecast from a consistently wrong one.

Q2.7

> For the mirror panel of video 9658 in Figure 3 (A, seed 0, 1x), compare rows 3 and 4 at h = 4 and 5. Starting from the identical mirrored inputs, direct prediction in row 3 draws the two digits together into a diffuse central patch by h = 5. Mirroring the original forecast in row 4 instead retains two displaced patches, with one above and to the right of the other. The difference is therefore in the forecast locations and merging of the digits, not just their reflected stroke shapes. The true future in row 2 shows the digits approaching and overlapping with recognisable strokes, so neither blurred account is a clean recovery of their motion and identity. This case has C_M = 0.00621 and E_M = 0.0230, compared with seed-0 mean C_M = 0.00172.

Q2.8

> Yes, for A at 1x rotation is harder on both measures in Table 3. Across seeds, C_R = 0.00280 versus C_M = 0.00178, about 58% larger, and E_R = 0.0236 versus E_M = 0.0226. Both transformed errors exceed the original E(5) = 0.0219. Figure 4 shows that the rotation distribution is broadly shifted to larger gaps, rather than the difference being caused only by the four selected outliers. For seed 0 the median gaps are 0.00261 for rotation and 0.00163 for mirroring.
>
> Q2.1 found that every direction is represented at normal speed, and Q2.2 explains that reflection and rotation preserve speeds and the square-box bounce rule. Rotation changes upright stroke orientation more strongly than mirroring, so dependence on digit appearance is a plausible explanation. The empirical direction histogram is not exactly symmetric, however, and these results alone do not isolate all effects of motion sampling from appearance. The A-to-B comparison will test this explanation by changing symmetry exposure while keeping architecture and training speeds fixed.

Faster motion

Table 4. Model A and Copy-last across speeds, $t = 1$. Only valid targets are scored, so the 3× column uses the two existing targets (frames 16 and 19).

Model	$E^{(2)}$ 1x	$E^{(2)}$ 2x	$E^{(2)}$ 3x	$E^{(5)}$ 1x	$E^{(5)}$ 2x
A	0.0185 (0.0188, 0.0182)	0.0455 (0.0447, 0.0463)	0.0534 (0.0526, 0.0541)	0.0219 (0.0223, 0.0216)	0.0501 (0.0484, 0.0517)
Copy-last	0.0548	0.0661	0.0703	0.0641	0.0719

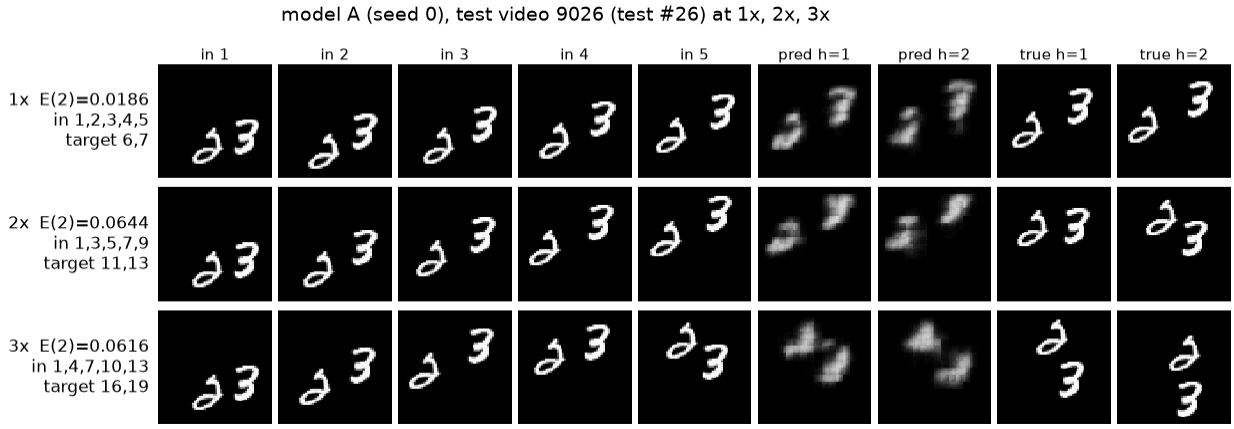


Figure 5. Model A (seed 0) on one test video whose digits do not overlap in any displayed true frame: five input frames, the first two predictions and the first two true frames at 1×, 2× and 3×. Test video 9026 (dataset index).

Q2.9

> The practical failure already appears at 2x. Table 4 gives mean A E(2) = 0.0185 at 1x, 0.0455 at 2x and 0.0534 at 3x, increases of about 2.46x and 2.88x over normal speed. A still beats Copy-last at each speed, but at 2x and 3x its E(2) is even worse than Blank (0.0425 and 0.0426 in the saved evaluation results). Its 2x E(5) rises to 0.0501 from 0.0219 at 1x.
>
> In Figure 5 (A seed 0, video 9026), normal-speed predictions are blurred but roughly located. At 2x, the predicted 3 stays near the upper edge while the true digit has rebounded downwards by the second target. The other digit also becomes a displaced smear. At 3x, both predictions are diffuse patches at positions that fail to follow the true bounced trajectories. These look like uncertain or incorrect extrapolation, with stroke identity lost, rather than a successful forecast made slightly less sharp. The per-video E(2) values are 0.0186, 0.0644 and 0.0616, so this selected case is slightly worse at 2x than 3x even though the test-set mean worsens with speed.
>
> Q2.3 predicts particular difficulty at 3x because the typical 9.95-pixel displacement exceeds the 8-pixel structural reach. Failure at 2x, where the typical displacement is 6.64 pixels, shows that this reach is an upper bound on dependence, not a guaranteed learned tracking speed. The estimate omits unfamiliar training displacements, effective use of the available paths, and the need to infer bounce timing and preserve a whole digit rather than one pixel.

Q2.10

> No. The A-only speed tests change the displacement distribution while also moving typical 3x motion beyond the calculated one-step reach, so the two explanations are confounded. Train C with A's unchanged architecture but uniformly sampled 1x/2x/3x windows. Improvement over A would show that unfamiliar motion mattered without any change in reach. Then compare C with D, trained with the same speed sampling, seeds and 30-epoch protocol, but with strictly larger reach. If C remains poor at 3x while D improves, that supports an architectural limitation after exposure has been controlled. D also changes parameterisation, so this is evidence for the reach explanation, not proof that reach alone determines the error.

4. What Can Data Fix?

Q2.11 (prediction, written and committed before training B, C and D)

> I predict that B will reduce the mirror and rotation failures of A, especially the larger rotation gap, by exposing the same motion to transformed digit appearances, but it will not guarantee exact consistency or repair the speed failure by itself. I predict that C will improve the 2x failure by learning from faster clips, while its unchanged 8-pixel one-step reach will leave the 3x clips difficult. I predict that D will improve 3x motion further through its larger reach and speed training, but without symmetry augmentation it may retain appearance-dependent mirror and rotation errors.

Inserted by the report build directly from the git commit recorded in `logs/phase3_gate.txt` (`answers/Q2_11.md`), committed at 2026-10-19T17:29:16+05:00 (commit fefd535). Training of B, C and the D candidates started at 2026-10-19T17:29:25+0500 (`logs/phase3_gate.txt`).

Model D selection (validation split only). Each candidate was trained exactly like a final model (30 epochs, seeds 0 and 1). The rule, fixed before any candidate was trained, is the lowest validation $E^{(2)}$ averaged over 1×, 2×, 3× and both seeds. The chosen candidate's two runs are model D. Entries are mean (seed 0, seed 1).

Additional experiment and chronology. The initial three-design comparison selected D_dil, whose seed 0 had a flat validation history and weak forecasts. After the original test analysis, I requested an improvement attempt on 20 October. One additional design, D_dil64, kept C's 64-channel width and changed only layer-2 dilation. Its two-run plan was committed as `8debfef` before training. The same validation score selected it at 0.0284, versus 0.0329 for D_dil, a 13.6% reduction. This selection was committed as `f4c92ad` before testing the new candidate. Both new seeds completed 30 epochs, bringing the total to 14 runs. The earlier test results had already been seen, so this is a disclosed adaptive extension, not an untouched-test comparison. The original runs remain available at commit `995521b` and the original selection in `results/d_selection_initial.json`. Q2.5 and Q2.11 are unchanged.

Candidate	Reach (px)	Params	val $E^{(5)}$ 1x	val $E^{(2)}$ 1x	val $E^{(2)}$ 2x	val $E^{(2)}$ 3x	mean val $E^{(2)}$ (selection score)
D_dil64 (chosen)	12	480,784	0.0269 (0.0263, 0.0276)	0.0241 (0.0235, 0.0246)	0.0290 (0.0286, 0.0293)	0.0323 (0.0319, 0.0327)	0.0284 (0.0280, 0.0289)
D_dil	12	703,888	0.0324 (0.0379, 0.0269)	0.0308 (0.0379, 0.0236)	0.0332 (0.0379, 0.0285)	0.0348 (0.0379, 0.0318)	0.0329 (0.0379, 0.0280)
D_k5	16	709,520	0.0344 (0.0395, 0.0293)	0.0327 (0.0395, 0.0260)	0.0343 (0.0395, 0.0291)	0.0356 (0.0395, 0.0317)	0.0342 (0.0395, 0.0289)
D_p8	16	695,584	0.0379 (0.0379, 0.0379)	0.0379 (0.0379, 0.0379)	0.0379 (0.0379, 0.0379)	0.0379 (0.0379, 0.0379)	0.0379 (0.0379, 0.0379)

Timing context. The D_k5 seed-1 screening run spanned a MacBook sleep/wake interval. The original training process continued after wake, and all raw epoch timer readings are retained. Power events and the interruption record are in `logs/sleep_wake_2026-10-19.txt` and `results/sleep_interruption.json`. Timing comparisons reflect these execution conditions and ordinary use of the Mac during the experiments.

The A, B, C and symmetrised-B measurements are retained from the original evaluation. The selected D and its wrapper were timed in the later session on the same Mac. Differences in elapsed time therefore also reflect session conditions, not only architecture.

Table 5. Models A to D on the test split. Training time is the mean wall-clock time of the 30 training passes, including window sampling and gradient updates. Validation and checkpoint writing occur outside that timer. Inference time is the time to predict the 1,000 original test videos (1×, $t = 1$) at batch 32 under `eval()` and `no_grad()`, timed after the batches are on the device, as the median of three full passes after three warm-up batches.

Model	$E^{(5)}$ 1x	C_M	E_M	C_R	E_R	$E^{(2)}$ 1x	$E^{(2)}$ 2x	$E^{(2)}$ 3x	Params	Train s/epoch	Inference s
A (baseline, 1x)	0.0219 (0.0223, 0.0216)	0.00178 (0.00172, 0.00183)	0.0226 (0.0229, 0.0222)	0.00280 (0.00271, 0.00290)	0.0236 (0.0238, 0.0234)	0.0185 (0.0188, 0.0182)	0.0455 (0.0447, 0.0463)	0.0534 (0.0526, 0.0541)	480,784	27.2 (26.6, 27.9)	1.12 (1.16, 1.08)
B (baseline, 1x, sym. aug.)	0.0229 (0.0237, 0.0221)	0.00139 (0.00143, 0.00135)	0.0229 (0.0237, 0.0221)	0.00166 (0.00160, 0.00173)	0.0229 (0.0237, 0.0221)	0.0191 (0.0198, 0.0184)	0.0453 (0.0453, 0.0453)	0.0523 (0.0521, 0.0525)	480,784	27.8 (27.1, 28.4)	1.21 (1.11, 1.30)
C (baseline, 1x/2x/3x)	0.0274 (0.0276, 0.0273)	9.26e-04 (6.27e-04, 0.00123)	0.0277 (0.0278, 0.0275)	0.00200 (0.00141, 0.00259)	0.0285 (0.0285, 0.0285)	0.0236 (0.0236, 0.0236)	0.0291 (0.0292, 0.0290)	0.0336 (0.0337, 0.0336)	480,784	31.2 (28.6, 33.8)	1.11 (1.09, 1.14)
D (D_dil64, 1x/2x/3x)	0.0272 (0.0265, 0.0279)	7.67e-04 (8.80e-04, 6.53e-04)	0.0274 (0.0268, 0.0280)	0.00202 (0.00213, 0.00192)	0.0284 (0.0278, 0.0290)	0.0243 (0.0237, 0.0249)	0.0291 (0.0287, 0.0295)	0.0329 (0.0325, 0.0333)	480,784	24.1 (22.9, 25.2)	0.466 (0.465, 0.467)

Q2.12

Model D is D_dil64, selected by the fixed validation score after the disclosed additional experiment. It keeps C's 4x4 pixel fold, two 64-channel ConvLSTM layers, 3x3 kernels, 1x1 output projection, pixel shuffle, sigmoid and nine-step autoregressive rollout. Only the second layer's dilation and padding change from 1 to 2. The first layer keeps dilation 1 and padding 1. D uses the same speed augmentation as C and no symmetry augmentation. With each seed, its initial parameter tensors exactly match C's because dilation changes the spatial connections without changing the weight shapes.

Under the Q2.3 conventions, one folded-grid cell is four original pixels. The longest path from the current input or previous first-layer hidden state passes through the first convolution's one-cell radius and the second convolution's two-cell radius. Reach is therefore (1 + 2) x 4 = 12 pixels. The previous second-layer hidden state has a two-cell path, or eight pixels. Pointwise gates, the output projection and reshuffles add no reach. The autograd check in logs/reach_D_dil64.txt confirms radii of three cells for u_tau and h1_prev and two for h2_prev. Thus D has strictly larger reach than A (12 > 8 pixels), with no attention.

D has the same 480,784 parameters as A and C: 184,576 in layer 1, 295,168 in layer 2 and 1,040 in the output projection, below the 721,176 limit. Both final-epoch candidate runs are reused. This gives a closer C-to-D comparison than the original 78-channel candidate, although dilation also changes which intermediate locations are sampled.

Q2.13

Appearance is the best-supported label for A's transformation-sensitive failure at 1x. In Table 5, changing only symmetry exposure from A to B lowers C_M from 0.00178 to 0.00139 (21.7%) and C_R from 0.00280 to 0.00166 (40.8%). Rotated error improves from 0.0236 to 0.0229, although mirrored error increases slightly from 0.0226 to 0.0229. B has almost no transformed-error penalty relative to its own original E(5) = 0.0229. This supports appearance dependence, especially under rotation, without showing that augmentation improves every forecast.

Unfamiliar motion explains a substantial part of the speed failure. C changes training speeds while retaining A's architecture and 8-pixel reach. Mean E(2) falls from A's 0.0455 to 0.0291 at 2x and from 0.0534 to 0.0336 at 3x, reductions of 36.0% and 37.0%. Exposure therefore produces a large gain without increasing reach.

Architectural reach receives modest additional support from the refined C-to-D comparison. D changes only layer-2 dilation, with the same width, initial parameter tensors, parameter count and speed training as C. Its mean 3x E(2) is 0.0329 versus C's 0.0336, a 2.11% reduction. Both seeds improve: D gives 0.0325 and 0.0333, versus C's 0.0337 and 0.0336. The 2x mean improves by only 0.231%, and the 1x first-two-horizon error worsens by 2.85%. Thus larger reach helps somewhat at 3x, but this is not the clean case of C failing while D succeeds. Dilation also changes the spatial sampling pattern, and two seeds do not establish statistical robustness.

The original 78-channel D_dil was worse than C on average because one seed plateaued. Its retained results show why reach alone does not guarantee useful learned dynamics. The additional 64-channel candidate improves mean 1x E(5) by 17.2% and 3x E(2) by 7.17% relative to that original D. This later candidate was selected on validation after the earlier test analysis, as disclosed above. Q2.11's prediction of further 3x improvement is supported only modestly by the final comparison. More training examples can improve computations along existing dependency paths, but cannot carry information beyond the architecture's one-step reach.

Q2.14

Yes. Relative to A's 1x E(5) = 0.0219, Table 5 gives B = 0.0229 (+4.46%), C = 0.0274 (+25.0%) and refined D = 0.0272 (+24.0%). B pays a modest original-speed cost for lower inconsistency and a small rotation-error gain, although mirroring is slightly less accurate. The trade is useful only when transformed appearances matter enough.

C and D offer a stronger trade for applications that must handle all three speeds. D reduces A's 2x and 3x E(2) by 36.1% and 38.3%. Against C, D improves 1x E(5) by 0.822% and 3x E(2) by 2.11%, but worsens 1x E(2) by 2.85%, so it does not dominate every metric. For original 1x clips alone, A remains more accurate.

D has the same 480,784 parameters as C. Its recorded training and inference times are 24.1 seconds per epoch and 0.466 seconds per 1,000 videos, versus C's 31.2 and 1.11 seconds. These are measurements from different sessions on the same Mac, so they do not establish a controlled speedup from dilation. The larger practical benefit of the refinement is avoiding the original D seed-0 plateau without increasing parameter count; the gain over the already useful C is small.

5. The Final Model

Why the symmetrised forecaster has $C_M = C_R = 0$. Every $g \in G$ only permutes pixels, so it is linear and commutes with averaging. For any $T \in G$,

$$\bar{f}(TX) = \frac{1}{8}\sum_{g \in G} g^{-1} f(gTX) = \frac{1}{8}\sum_{g' \in G} Tg'^{-1} f(g'X) = T\bar{f}(X),$$

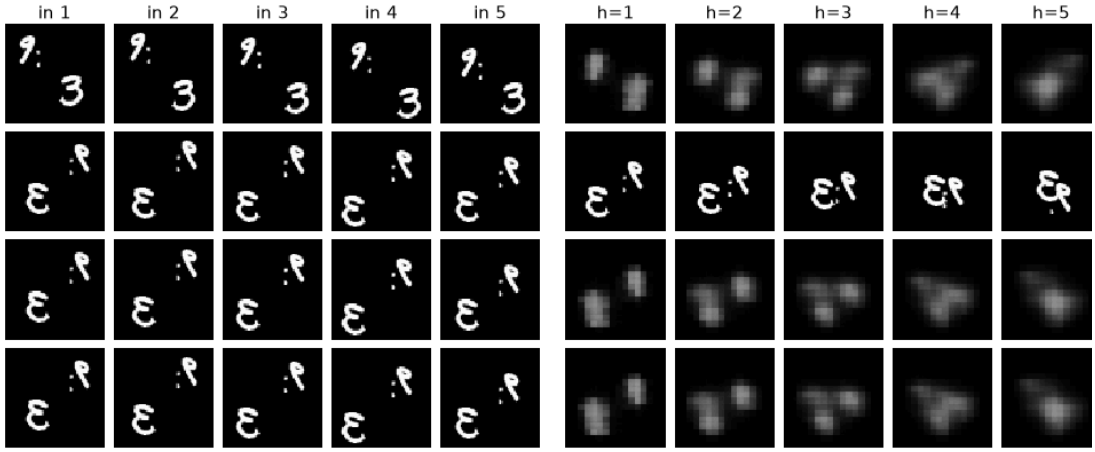
substituting $g' = gT$ (so $g^{-1} = Tg'^{-1}$), which runs over all of G as g does. Hence $\bar{f}(TX) = T\bar{f}(X)$ for every X, and $C_T = MSE(\bar{f}(TX), T\bar{f}(X)) = 0$ for $T = M$ and $T = R$, whatever f is. In floating point the eight terms are summed in a different order on each side, so the measured gaps are at rounding level.

Final choice: symmetrised D (D_dil64). Both validation-selected D checkpoints are wrapped without further training. I did not add optional D+B. In Table 5, B reduces inconsistency and improves rotation error modestly relative to A, but slightly worsens original and mirrored errors. Refined D has some remaining rotation sensitivity, yet the table does not establish that adding B's augmentation would improve its overall accuracy. I therefore use the prescribed symmetry wrapper and retain the completed controlled comparison. The renewed decision and the updated Table 5 hash are in results/db_decision.json.

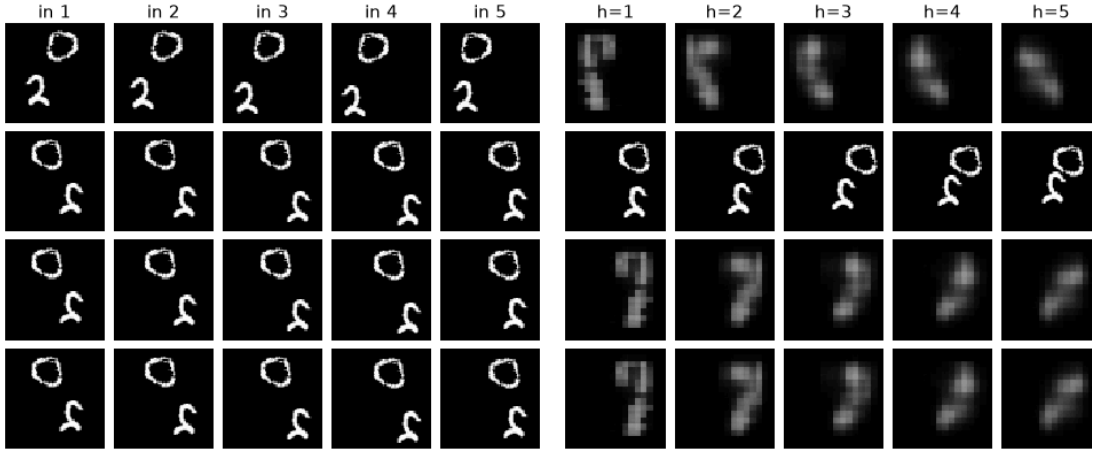
Table 6. Symmetrised models, same columns as Table 5. Training time is that of the underlying learned model. Inference time includes all eight transformations, forward passes, inverse transformations and the averaging.

Model	$E^{(5)}$ 1x	C_M	E_M	C_R	E_R	$E^{(2)}$ 1x	$E^{(2)}$ 2x	$E^{(2)}$ 3x	Params	Train s/epoch	Inference s
symmetrised B	0.0222 (0.0231, 0.0214)	9.88e-17 (9.87e-17, 9.89e-17)	0.0222 (0.0231, 0.0214)	8.00e-17 (8.04e-17, 7.96e-17)	0.0222 (0.0231, 0.0214)	0.0187 (0.0194, 0.0179)	0.0447 (0.0448, 0.0446)	0.0513 (0.0513, 0.0514)	480,784	27.8 (27.1, 28.4)	10.2 (9.45, 10.9)
symmetrised D (D_dil64) = final model	0.0272 (0.0265, 0.0279)	6.70e-17 (6.60e-17, 6.80e-17)	0.0272 (0.0265, 0.0279)	5.30e-17 (5.34e-17, 5.26e-17)	0.0272 (0.0265, 0.0279)	0.0243 (0.0238, 0.0249)	0.0291 (0.0287, 0.0295)	0.0328 (0.0324, 0.0333)	480,784	24.1 (22.9, 25.2)	3.84 (3.84, 3.84)

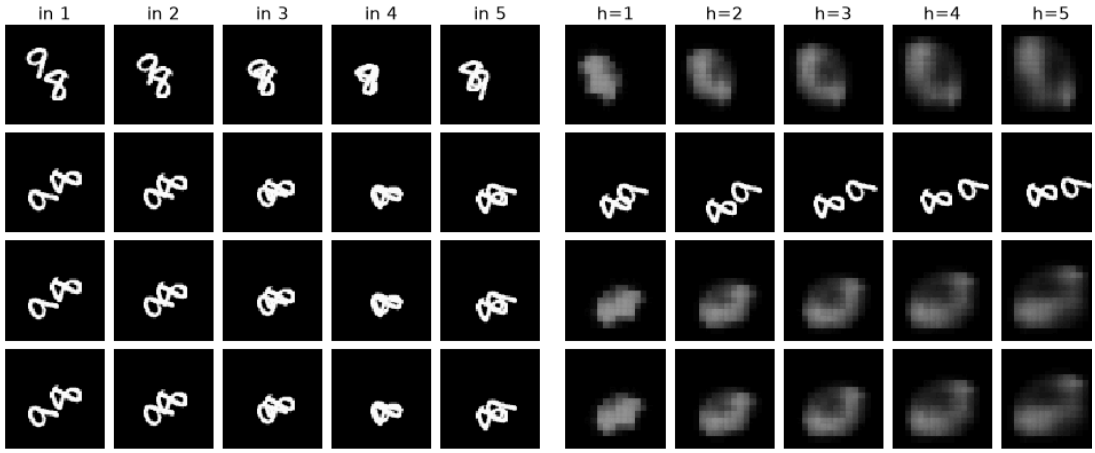
final model: symmetrised D (D_dil64, seed 0), T = M, test video 9658 (test #658): per-video C_M = 7.95e-17, E_M = 0.0292



final model: symmetrised D (D_dil64, seed 0), T = M, test video 9257 (test #257): per-video C_M = 8.17e-17, E_M = 0.0321



final model: symmetrised D (D_dil64, seed 0), T = R, test video 9279 (test #279): per-video C_R = 9.56e-17, E_R = 0.0389



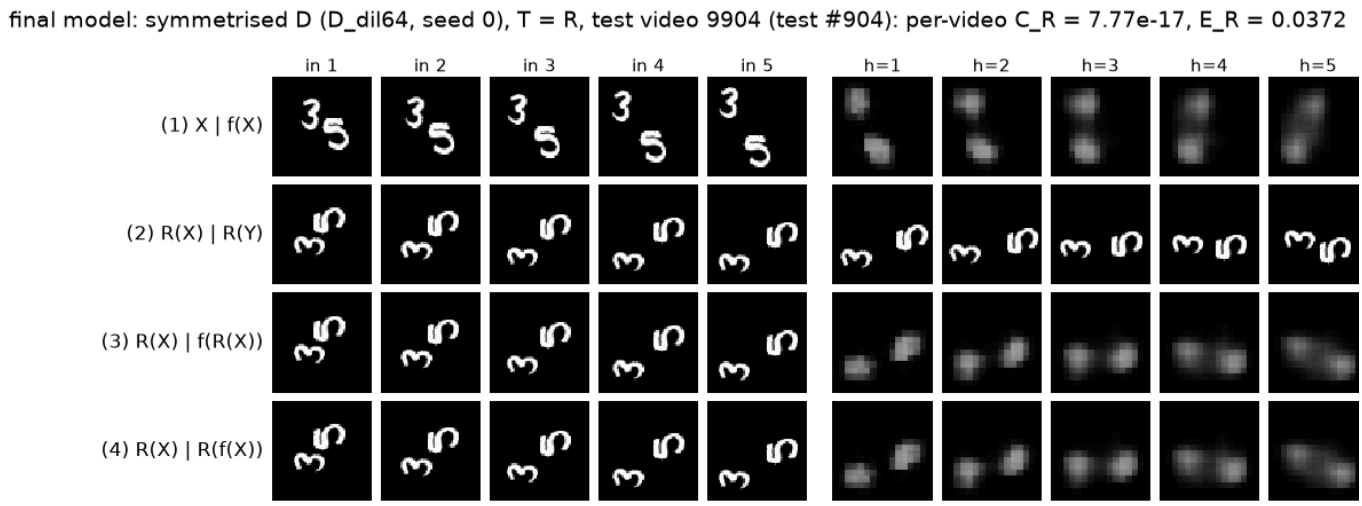


Figure 6(a). Figure 3 repeated with the final model on exactly the same videos. GIFs: `gifs/fig6a_final_M_v658.gif`, `gifs/fig6a_final_M_v257.gif`, `gifs/fig6a_final_R_v279.gif`, `gifs/fig6a_final_R_v904.gif`.



Figure 6(b). Figure 5 repeated with the final model on exactly the same video.

Q2.15

> Learned symmetry in B is approximate. Table 5 gives C_M = 0.00139 and C_R = 0.00166, alongside E_M = E_R = 0.0229 after rounding. Original E(5) is also 0.0229. Relative to A, the consistency gaps are smaller, but they remain nonzero and original-speed accuracy is slightly worse.
>
> Group averaging is exact in real arithmetic because its eight terms can be reindexed under any square symmetry, as proved above. In Table 6, symmetrised B has C_M = 9.88e-17 and C_R = 8.00e-17, with original and transformed errors all 0.0222. Final symmetrised D has C_M = 6.70e-17 and C_R = 5.30e-17, with corresponding errors 0.0272. These gaps are floating-point residuals. Consistency is guaranteed, while correctness still depends on the learned forecasts.
>
> Wrapping B improves original E(5) by 2.97% and improves both transformed errors. For D, wrapping slightly worsens original E(5) by 0.0703% before rounding, with small increases for both seeds. Its 1x and 2x E(2) also increase by about 0.15%, while 3x E(2) improves by 0.268%. Mirrored and rotated errors improve from 0.0274 and 0.0284 to 0.0272. The wrapper therefore buys exact agreement and transformed-scene accuracy at a very small original-scene cost.
>
> Inference rises from 1.21 to 10.2 seconds per 1,000 videos for B (8.43x), and from 0.466 to 3.84 seconds for D (8.23x). Parameter count and underlying training time are unchanged. Averaging can blur inverse-aligned forecasts that disagree in position or shape, and hurt an originally accurate orientation when other views are worse. Mean squared error averaged over the whole symmetry orbit cannot increase relative to the corresponding average base-model risk, but error on the original orientation can increase. D exhibits that small original-speed cost here; its Figure 6 predictions also remain visibly blurred.

Q2.16

> The mirror disagreement is fixed, but the forecast is still not fully correct. For video 9658 at 1x, Figure 6(a) rows 3 and 4 now agree: C_M is 7.95e-17 versus A's 0.00621. The refined final seed-0 model predicts two moving patches that approach and merge, rather than the featureless output of the original D candidate. However, the 9 and 3 lose their strokes, and the overlap becomes a diffuse mass. Its E_M = 0.0292 is still worse than A's 0.0230 on this case, although much better than the original final model's 0.0451. Consistent motion-like output is an improvement, not a successful reconstruction of both digits.
>
> Group averaging fixes commutation exactly. The A-to-C results attribute most of the population-level fast-motion gain to speed exposure; the C-to-refined-D comparison supports a smaller additional 3x gain from the changed spatial connections. These comparisons do not isolate each change's effect on this individual video. In Figure 6(b), video 9026 now has localized but blurred forecasts, with E(2) = 0.0271, 0.0374 and 0.0388 at 1x, 2x and 3x. The fast-speed errors improve on A's 0.0644 and 0.0616, but the normal-speed error exceeds A's 0.0186. At faster speeds, the patches still lose digit shape and miss precise positions and bounce timing.

Q2.17 (final verdict, at most 150 words)

> The final model learned useful predictive structure, but a general motion rule is not established. Speed exposure accounts for most of the improvement: C reduces A's fast-motion errors by about 36-37%. Refined D improves 3x error by a further 2.11% over C, with both seeds benefiting, and its wrapper guarantees symmetry consistency. Its forecasts now move and localize the digits, but remain blurred and unreliable through overlap and bounces. Original-speed error still exceeds A's. A skeptic would need reliable trajectories and wall bounces across speeds, shapes and seeds, rather than consistency or background-dominated pixel MSE alone. One missing test is forecasting unfamiliar non-digit shapes with known trajectories. The additional D design followed earlier test analysis, so an untouched evaluation set would also strengthen the evidence.

AI-use disclosure

Tools, outputs and roles. Claude (Anthropic) supplied the initial Q2 implementation, experiment scripts, Q1 proposal and report drafts. Codex (OpenAI) reviewed that work against the handout, corrected implementation and reporting issues, executed the experiments, analysed results and checked the PDF layout. Execution and review evidence is in `WORK_LOG.md` and `reviews/`. Measurements were produced locally on Apple M1 Max, 64 GB RAM, macOS-15.6.1-arm64-arm-64bit-Mach-O, PyTorch 2.14.1, device mps.

Reconstructed prompt sequence. These prompts were written retrospectively to express the instructions and review questions for each stage. They are forward-looking examples, not a verbatim record of the original interactions or evidence that these exact prompts were executed. The actual work produced and subsequent edits are summarized separately below.

Step 1. Establish the review scope

> Review the handout and supplied implementation before changing anything. The central question is whether the predictor learns transferable motion or relies on familiar appearances and displacements. Map each requirement to code, evidence and report sections. Separate mandatory constraints from design choices, identify unsupported claims, and propose a staged review with the two prediction-commit gates.

Step 2. Challenge the river-forecasting proposal

> Review Q1 as a forecasting-system proposal. Distinguish supplied facts from assumptions, and explain how each assumption changes the design. Trace timestamps, tensor dimensions, spatial paths and temporal reach through the architecture. Check that inputs were available at issuance, address missing and delayed records separately, and challenge unnecessary complexity or physical claims the data cannot support.

Step 3. Validate data, transformations and scoring

> Audit complete-video splitting, scaling, window indices and augmentation using small cases whose answers can be checked by hand. Derive valid starts at each speed and verify all five predictions are generated even when fewer targets exist. Check equal-video loss weighting, transformation inverses and metric aggregation. Return failing examples and focused fixes before any training.

Step 4. Review and run the baseline

> Check the baseline against the specified equations, default initialization and nine-step rollout. Derive its parameter count and longest within-update path in original pixels, and verify gradients through feedback. Once checks pass, train both seeds with the fixed protocol and inspect normal-speed results. Prepare Q2.5 from that evidence and commit it before running any stress test.

Step 5. Diagnose failures before changing the model

> Run the prescribed mirror, rotation and speed tests. Pair consistency gaps with transformed errors and inspect trajectories, overlap and bounces in the required examples. Explain which observations support appearance dependence, unfamiliar speed or limited reach, and which remain confounded. Form a Q2.11 prediction in at most three sentences and commit it before training B, C or D.

Reconstructed prompt sequence, continued

Step 6. Run controlled comparisons

> Keep B and C identical to A except for their prescribed augmentations. Compare defensible D architectures with larger reach and within the parameter budget, using the same training protocol. Fix the validation selection rule before training, retain both seeds and all candidates, and reuse the winner's runs. Interpret Table 5 without treating increased reach as guaranteed improvement.

Step 7. Investigate weak D performance

> Inspect both seeds' training and validation histories before proposing a remedy for the weak D result. Compare optimization failure, width changes and spatial connectivity as explanations. Plan at most one additional candidate that keeps C's width and increases dilation. Record the plan and prior test exposure, retain the full protocol, and commit validation selection before evaluating the new candidate on test data.

Step 8. Evaluate the final symmetry wrapper

> Derive why inverse-transforming and averaging all eight group predictions guarantees equivariance. Verify the implementation and report consistency, transformed accuracy and inference cost separately. Decide optional D+B from Table 5 evidence, then repeat the original failure examples with the final model. Explain remaining blur and errors, and whether the evidence establishes motion learning or only improved prediction.

Step 9. Audit the submission

> Check every answer, table, figure, caption and GIF against the handout and saved results. Preserve the original prediction text and both seeds, and identify unsupported causal claims or missing requirements. Verify reproduction instructions and the rendered PDF. Summarize AI-produced work and subsequent corrections accurately, distinguishing documented interactions from reconstructed guidance, and report unresolved issues before calling the submission ready.

Produced work, edits and verification.

Work produced. AI assistance produced the implementation and experiment scripts, the Q1 proposal, measured-result analysis, tables, figures, GIFs and report drafts. The initial and subsequent tool roles are identified above; code, logs and reviews provide the corresponding artifacts.

Subsequent edits and checks. Codex corrected package setup, added and tested per-epoch recovery, corrected report assembly, numerical formatting and GIF timing, and revised Q1 assumptions and data-availability claims. Window, transformation, gradient and reach checks supported implementation review. Reported measurements retain both seeds, the original prediction commits and the disclosed later D experiment.